

# A Web Platform for Recipe Scaling and Per-Serving Ingredient Delivery with AI-Powered Cooking Assistance

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**Abstract**—Home cooking can cause many challenges, including the purchase of more than what is needed or an inaccurate measurement of the ingredients, which later results in food wastage and inconvenience. This has been observed to be an especially problematic area among university students and working professionals in urban localities, who might not have ample time to shop and to prepare meals in the most efficient way possible. In solving this issue, we would like to suggest a scheme of the Smart Ingredient Delivery System that is an online platform allowing to choose the number of servings and generate the appropriate amounts of ingredients in response to this choice. The system is designed in a modular structure, and its implementation is based on a React interface, a server implemented with Node.js/Express and a PostgreSQL database. Its key characteristics are: (1) automated scaling option to recalculate the amounts of the ingredients, (2) a customizing interface where it is possible to replace the ingredients and change the level of spices, and (3) an artificially intelligent chatbot based on the information about recipes to give the cooking instructions and troubleshooting advantages. We also introduce the prototype of the system and an evaluation plan that regards not only the volume of food waste saved in contrast to meal-kit stores but also correctness of the chatbot responses. All in all, the system is supposed to offer a family friendly and environmentally friendly way of cooking at home with accuracy, customization, and AI-based support.

**Keywords**—*ingredient scaling, meal-kit delivery, AI chatbot, food waste reduction, recipe customization*

## I. INTRODUCTION

The topic of household food waste is still an urgent problem in the world, and overbuying and the incorrect use of ingredients in a home-cook meal are also considerable contributors to this problem [1]. The challenges are being specifically acute in urban places, with busy professionals and even students who have little time to go shopping in the

grocery and cook the meal, which leads to more wasteful choices [2].

Traditional food delivery services and meal box companies, like Zomato, Swiggy, and Blue Apron, have tried to mitigate those problems by enhancing the accessibility of ingredients and convenience. Such services however tend to be based on pre-bundled or percentage-based models, which are not always based on the needs of individual households and they often result in excess of ingredients [3]. The study of adoption of meal-kits further points out that, despite convenience, fixed-portion systems do not allow flexibility and customization, which is constraining their ability to minimize waste [4].

The recent development in the field of artificial intelligence (AI) creates a chance to eliminate these shortcomings. Smart recipe recommender and cook assist systems are more and more able to adapt food to personal needs and tastes [5]. Nevertheless, all these solutions are divided, and some are concentrated on the recommendation aspect, some on ingredient substitution, and some on chatbot-based cooking support, but none of them incorporates all functions onto a single platform. To fill this gap, this paper introduces a web-based application that will deliver the Smart Ingredient Delivery System which aims to minimize the amount of food wasted by the household and increase the convenience of users. The system promotes scaling of ingredients serving, dynamism of recipes customization and order management and introduces an AI-based chatbot component to assist cooking content and troubleshooting. The viability of this solution has been proven with a prototype implementation which shows how recipe search, scaling of ingredients, customization, and order processing can be integrated into a scalable platform.

## II. RELATED WORK

The existing articles on the question of food sustainability, personalization, and AI-based cooking assistance have grown in various complementary directions that form the assumption of the Smart Ingredient Delivery System. The

first study of intelligent food recommendation system was to formulate more precise utilization of contextual information and user inclinations and ingredient level presentations to amplify the accuracy of recipe browsing [6]. It was further perfected by the use of ingredient embedding techniques that coded semantic relationships between ingredients that led to adaptive ingredient replacement and the identification of similarity that helped in the custom recipe generation [7]. These models were the basis of the culinary recommender system scalability to the personalization of the culinary.

Other studies have been conducted in the same regard to reduce food wastage through technology and behavior based interventions. In an in-depth examination, it has been mentioned that portion control, data-driven cooking appliances, and education could significantly reduce the quantity of food waste at the domestic level, yet the majority of the solutions were not incorporated into the consumer-level frameworks [8]. This gap had brought to the fore the necessity to come up with integrated solutions that would streamline recipe choice, portioning, and preparing into a seamless ecosystem.

The addition of artificial intelligence to the cooking assistant has introduced a new level of interactivity and automation of functions. The Michigan KitchenScale was a system, which employed AI-based ingredient prediction models to endeavor to forecast the amounts of ingredients in a dynamic manner, depending on the context of a recipe and lessening erroneous manual ingredient quantification [9]. On another note, the development of smart cooking assistants that apply AI to guide users with real-time advice on recipes preparation was developed [10]. Though these studies developed a more convenient user experience, their main objective was not to develop the delivery logistics integration and control of ingredients. Prolonged literature reviews of food recommender systems have discovered gaps that portray absent but prominent concerns, such as the lack of sparseness of data, explainability, and user trust [11]. At the same time, the innovations such as NutriScan proved the prospects of computer vision and AI to recognize and analyse ingredients on pictures, which is inclined to the growing popularity of smarter ingredient analysis [12]. These models are however used in the analysis and not in logistics of preparing and delivering meals.

Conversational or rule-based recipe assistants have also been strived to be developed. The Smart Recipe Assistant was such a system based on simple natural language processing and procedural logic to prepare a recipe with the help of a robot, which improved time efficiency of cooking, but did not have dynamically scaled ingredients and no price-setting algorithms [13]. The consumer behavior and sustainability studies also revealed that the impressions of the packing waste considerably influence the meal-kit introduction since the users are eager to find a convenient solution but are concerned about the environmental footprint [14]. These lessons support the need to develop systems that can be utilized and sustainable in terms of reduction of wastes.

More recent systematic reviews of food recommender systems have also found the existence of algorithmic approaches such as collaborative filtering, content-based, and

hybrid approaches, which also indicate the research gaps in the area of AI-based personalization combined with supply chain management [15].

The reviewed literature indicates that the isolated areas primarily have achieved the progress of personalized recommendations, AI-based cooking help, and sustainable meal delivery, yet it has still not manifested in all areas. The existing systems are either personalization of recipes, intelligent assistance, or ingredient logistics separately. The Smart Ingredient Delivery System is the solution that treats all of these areas in one consistent system, scaling ingredients, customizing recipes, tracking orders in real time, intelligent assistance all in one system which makes it a complete approach to overcoming all the limitations of the former studies.

### III. SYSTEM ARCHITECTURE AND METHODOLOGY

Smart Ingredient Delivery System proposed here has a goal to maximize home cooking via per-serving, custom ingredient delivery with AI-driven cooking assistance. It integrates a modular backend, frontend, database, and domain-specific AI chatbot. Details regarding methodology specify operational workflow, algorithms, as well as evaluation tactic.

#### 3.1. System Architecture

Smart Ingredient Delivery System is created as a web application that can be referred to as modular because it possesses a defined front end, a defined backend, a defined database and AI modules. The overall architecture is described in Fig. 1 which presents the interaction of the user interface, the backend services, databases, and AI-driven modules. The frontend is a one-page app, which has been created using React to achieve an interactive and responsive interface. It enables management of user accounts and profile, the possibility to view recipes and have details view, option to replace or remove ingredients, case-by-case adjustment of spices and shopping cart and in-built chatbot features to give real-time cooking instructions.

The backend will be implemented as a RESTful server, an Express server, running on Node.js, which will be the mediator between the frontend and databases and the AI services. It is modeled as a series of service layers that have modules where user service is in charge of handling the login, registration, and storage of preferences, recipe service is in charge of CRUD on recipes, scaling of the serving-size, order service and inventory service in charge of ingredient availability, order check out and tracking of the orders, chatbot service is in charge of the processing of user queries, intent mapping and writing of the response.

The database engine is the PostgreSQL which the structured data such as user information, recipes, information about ingredients and order history are stored. The schema is created in a way that is scalable and self customizable; it is created with schema Recipes (id, title, steps, base-servings, metadata), Ingredients (id, name, unit, cost-per-unit, inventory-count), RecipeIngredients (recipe-id, ingredient-id, quantity-per-serving), Orders, with OrderItems, where purchases are processed and ingredients are allocated.

The chatbot service is a domain-specific natural language model based on the AI trained on the instructions on recipes, and the ingredients. It further provides the option of real time cooking assistance, replacement of ingredients, and customization of serving sizes and addresses most frequent queries related to cooking and makes the user more interactive.

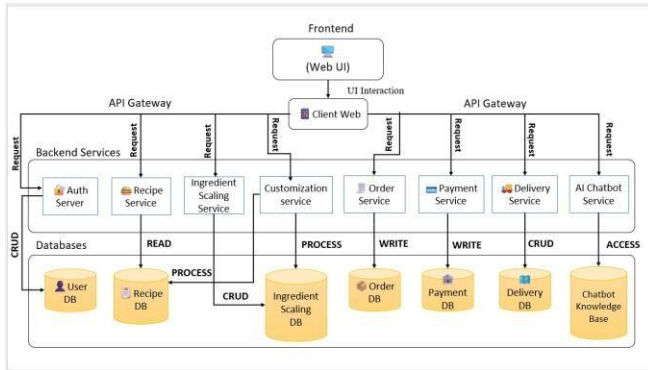


Fig:1 Architecture diagram

The system architecture is shown in Fig. 1. The frontend is connected to the backend services via API gateway and provides communication that is secure and well structured. Auth Server handles login and user preferences and Recipe Service retrieves and updates recipes. Ingredient Scaling Service refers to a dynamically changing quantity of ingredients according to the size of the serving and is connected to the Ingredient Scaling Database. An ingredient swap or removal is a Customization Service. The Order and Payment Services service handles orders, cost calculations, and transactions and the Delivery Service updates the logistics status. Lastly, the AI Chatbot Service that has been supported with a knowledge base will offer conversational cooking support and recipe troubleshooting.

### 3.2. Methodology

The approach to the proposed Smart Ingredient Delivery System provides an explanation to the design and implementation of the main modules of the approach-ingredients scaling, recipe modification, order-handling, and a computer-aided interaction. The system has a modular architecture that connects the frontend and the backend using the RESTful API to PostgreSQL to maintain data consistency. All of the components cooperate to offer meal personalization, efficiency, and decrease in food waste.

#### 3.2.1. Ingredient Scaling Algorithm.

The ingredient scaling algorithm is the primary component of the personalization engine of the Smart Ingredient Delivery System. It makes relevant calculations in regard to the quantity of ingredients to be used based on the number of servings indicated by the user. This ensures that the correct proportionality and elimination of excess foods waste which is normally occasioned by oversized buying or poor forecasting is avoided. The algorithm gives the conversion of the entire unit conversion, smooth conversion of gram and cups and table spoons among other common measures. This is what provides the user with a convenient and flexible

interface in which one can visualize the amount of ingredients in the system of their preference.

To further enhance the accuracy of the operations, the algorithm incorporates a rounding mechanism in the calculated quantities in to ordinary units of purchase that will ensure the procurement of the ingredients is in line with the actual packaging and retail features of the world. Moreover, it has a good database transaction system that enforces atomic ingredient bookings. This ensures consistency in inventory and prevents conflict such as over selling and allocation. All the reservations are executed independently as a separate transaction and this guarantees that integrity and transparency of all orders compared to the entire inventory system is maintained.

#### 3.2.2. Customization of Recipes Workflow.

The interactive customization of recipes within the working process is one of the aspects of divergence of the proposed system. This is due to the fact that the users have a chance of modifying the offered recipes to suit their diet, cultural inclinations or access to foods. Having an interactive user interface, users could add, delete and substitute ingredients in real time without having to disrupt the format of the dish or affecting the food taste. The site also provides alternatives on the various degrees of taste i.e. degree of spiciness or sweetness such that the recipes may be customized to individual taste in cooking.

All the customization would be processed in real-time in the back-end system and compute the re-pricing of ingredients and determining the prices based on the replenishing of ingredients. This active redefinition increases the consumer experience and also ensures that the ingredient acquisition experience is brought to the current availability and price. The conclusion is a flexible and human adaptable cooking helper which may offer flexibility and precision.

#### 3.2.3. Processing and Checkout of Orders.

Through the checkout, the system will perform an extensive check and availability of all ingredients ordered. This ensures that the ingredients required to prepare the recipe are in place hence avoids the risks of over selling or failing to meet orders. It is also associated with secure payment gateways like Stripe and RazorPay to make and receive financial transactions safely and efficiently. Such integrations facilitate multi currency payments, encryption and automatic confirmation of payment of sensitive data to improve reliability and trustworthiness.

In the case of successful order, a system can generate precise packing lists, purchase summaries. They are products that simplify the order preparing process and ensure that the suppliers and delivery partners receive proper and organized instructions. Consequently, this module is applicable in making sure that the operational efficiency is improved, and that the customers receive their ingredients promptly, and in sufficient amounts.

#### 3.2.4. AI Chatbot Integration

A chatbot based on AI will be incorporated into the Smart Ingredient Delivery System, and it will serve as an interactive cooking assistant. The chat bot is founded on natural

language processing (NLP) system which allows the chatbot to read and respond to queries of the user in an asynchronous and contextual way. The chatbot will give extremely precise and contextual answers to the cooking circumstances of the user based on organized information in recipes, libraries of ingredients and scaling algorithms.

Besides giving solutions to the general questions, the chatbot also helps the user throughout the cooking. It provides instructions stage by stage, offers alternative ideas, explains how-to prepare, and customizes its responses to the current point of interaction with the user. Such dynamic and intelligent communication is an enormous addition to the user interaction and satisfaction which makes the cooking process a collaboration between the human and machine intelligence.

This conversation AI agent can bridge the disconnect between the archiving of recipes and the active culinary recommendation, and with an active ecosystem, more accessibility, personalization, and convenience can be achieved in everyday cooking.

#### IV. PROTOTYPE IMPLEMENTATION

A prototype of one of the elements (recipe personalization) was successfully tested through the Smart Ingredient Delivery System to verify the viability and compatibility of the scaling of ingredients and intelligent management of orders within one system. It was a web application based system on three tier design presentation, application logic and data storage.

The frontend was also created with React.js that allows creating the user interface that is intuitive and responsive. It adheres to the component-based design whereby the separate functional units, such as the recipe search, customization, cart management, and order placement, are modeled in each reusable React component. This modularity simplifies the development process in addition to scalability and maintainability of the process. The frontend is dynamically displayed depending on the user interaction through React state management, including real-time feedback response to the action like ingredient deselection, scaling the serving and updating the price.

Its back-end layer was created using the Node.js and express, and is in turn founded upon the services of the RESTful API which process user requests and transfer of data. This type of APIs deals with some basic functions of the system such as finding recipes, scaling ingredients, user authentication and taking orders. The back-end will be linked to the PostgreSQL database and the AI microservice via asynchronous requests, which will make it responsive and efficient, especially with customer loads.

In order to obtain data persistence, A PostgreSQL relational database was employed since it is robust, satisfies the requirements of the ACID, and scaled. The schema design will be made up of User, Recipe, Ingredient, RecipeIngredient, Order and OrderItem relational tables. With the help of such relations, the efficient relationship will exist between recipes and the corresponding ingredients and will specify the proper scaling and costing of dishes of various sizes of serving. This schema also allows the user specific data to be stored and this is made easy through the

tracking preferences and personalized recommendations in subsequent iterations.

The chatbot assistant is an AI, which was supposed to be an interactive cooking assistant. The chatbot is developed on the premise of a natural language processing (NLP) model, which analyzes the user requests in terms of ingredient replacements, recipes, and nutrition. The chatbot architecture allows the application to make calls to the backend APIs and get contextually relevant information (e.g. recipe directions or ingredient conversions), and even send real time conversational responses.

In conjunction with development, functional testing was also done on an aggressive level to test all the modules. The testing scenarios were accuracy in recipe search, dynamic scaling behavior, correct ingredient selection, inventory validation on checkout and successful order confirmation process. The results supported the premise that all of the components in the system interact harmoniously across various levels and therefore, pointed to the fact that the proposed system can be viable.

#### V. RESULT AND DISCUSSION

Smart Ingredient Delivery System proves to be an innovative and efficient product that is meant to revolutionize the process of treating food and meal preparation in the home of the user. The system also helps the users order exactly what they need by combining smart ingredient measurement, personalized recipes and automated ordering capabilities, which will greatly reduce food wastage. Its easy to use interface encourages convenience as it enables the user to adjust the size of portions, the ingredients used, and changes in nutrition and costs are instantly shown. Moreover, the ability to integrate with grocery websites easily makes the ordering process smooth and deliver all fresh products on time. On the whole, the prototype focuses on a sustainable vision, including the technologies, user experience, and environmental friendliness to improve contemporary cooking habits.

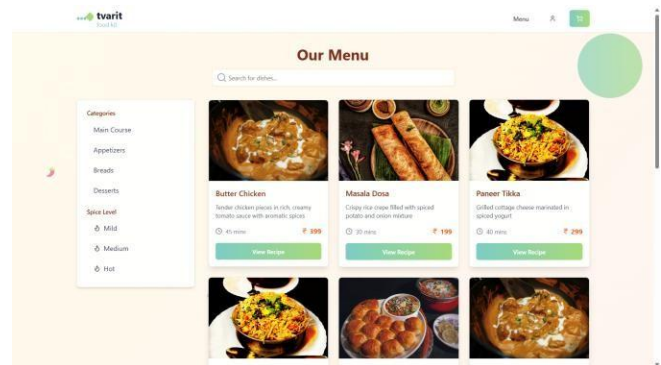


Fig. 2. Search Interface and Menu Browsing.

The menu browsing feature gives its users an opportunity to navigate through the different types of meals based on the course as indicated in Fig. 2. Live search and filtering system has instant suggestions, which increases discoverability and user satisfaction. The system design has been developed to promote exploration and enable users to pick the food according to their preferences, eliminating the use of impulse and poor decision-making when placing orders, which is a primary reason to end up with food wasted.



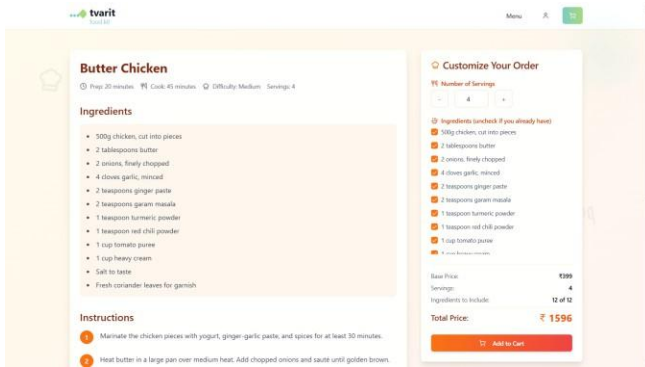


Fig. 3. Recipe Customization Interface and Scaling Interface.

The recipe customization module is very interactive and adaptive as indicated in Fig. 3. Customers are given the option of changing the serving size, discarding ingredients that they already have, or the intensity of flavor. The calculation of the ingredients quantities and their respective costs is automatically and dynamically recalculated in real-time allowing it to be precise and transparent. This interactive process improves customer control, reduces unnecessary buying and facilitates sustainable consumption habits.

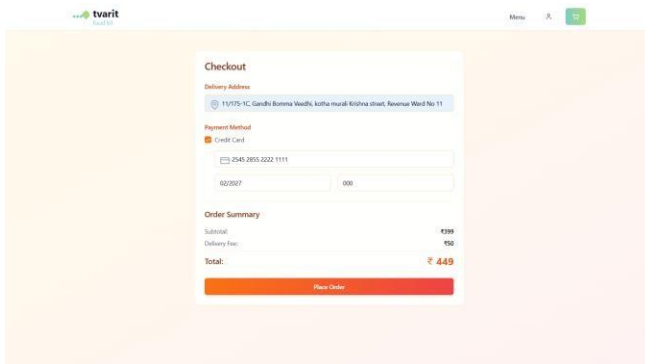


Fig. 4. Order placement interface and checkout interface.

The last user experience is the checkout and order placement interface as indicated in Fig. 4. The clients will be able to check their cart, confirm the quantity of ingredients used and make payments through secure methods like Stripe and RazorPay, which are already integrated into the platform. The order summary generated dynamically gives a clear view of the total cost which is updated by customization. This automation guarantees accuracy in the running of operations and the orders are served with minimal human involvement. Performance wise, the prototype demonstrated quick response to user actions, performance database retrievals, and real time updates on the system with a number of concurrent users. The smooth frontend-backend layer interconnection reflects the scalability of the platform and the availability of the expansion to the production level deployment.

According to the discussion of results, the system does not only optimize ingredients and personalize meals, but it also offers interactive and educational cooking experience. The design of AI chatbot (still being developed) is supposed to help close the gap further between the unchanging databases of recipes and the real-time support, which will eventually provide improved user interaction and confidence in cooking. Overall, the findings confirm the capacity of the

system to integrate numerous technological and behavioral elements such as scaling accuracy, personalization, and interactivity into an all-inclusive and functioning system that meets the needs of consumers and environmental sustainability.

## VI. CONCLUSION AND FUTURE WORK

we presented the Smart Ingredient Delivery System, a web-based platform created to eliminate home cooking issues of overpurchase, incorrect ingredient measurement, and lack of recipe customization. Using integration of ingredient scaling per serving, real-time recipe customization, and module-based order management, the system enables access to ingredients properly measured to order. Prototype development demonstrated principal functionality including recipe selection, ingredient scaling, customization, and order processing enabled by React.js frontend, Node.js/Express backend, and PostgreSQL database. It supplies a more flexible solution to typical pre-fixed meal-kit services by supporting dynamic customization and reducing potential waste related to meals. Additionally, its module-based construction is capable of supporting scalability as well as integration with higher-end features.

For future development, integration of the AI-based chatbot module will be complete to enable interactive support for cooking as well as troubleshooting. Other additions include real-time tracking of delivery via courier GPS-based APIs, incorporation of nutritional analysis as well as allergy identification, and addition of a voice-based recipe assistant. Enlarging the system to accommodate multi-lingual as well as multi-cultural recipes is also a major area of development. These additions will increase the system's ability to enhance convenience, minimize waste, as well as enhance sustainable usage for urban consumers.

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